Data Inspection for Each Table

Background:

This document describes the main findings about the data before data cleansing. It also explains the data definitions and provides basic metadata for each field across all tables. All counts were verified against the source data.

Loading note. The CSV files were loaded into BigQuery with schema auto-detection. Empty CSV fields therefore became NULL (no blank strings exist in any column), and phone in customers and dealers was typed INT64, so its leading zero was lost (the CSV files themselves are intact).

Dataset: customers 10,690 rows (9,742 Retail, 724 Corporate, 224 Fleet); dealers 32 rows; vehicles 62 rows; transactions 11,939 rows (2021-01-01 to 2026-09-27).

Summary of findings

Table / field	Finding	Handling
customers / full_name	84 Retail customers are registered twice (same name, phone and birth date); 43 Retail names are in capital letters.	master_customer_id (soft merge): 10,690 records = 10,606 unique customers. INITCAP(TRIM()).
customers / phone; dealers / phone	Typed INT64 (leading 0 lost). Customers: 10,388 start with 8, 24 with 62, 278 NULL. Dealers: all 32 have 9 digits.	Prefix 62 to numbers that start with 8 (all dealer numbers).
customers / city	149 spellings for 36 cities (capitalization, leading and trailing spaces); 330 rows.	INITCAP(TRIM()).
NULL, not applicable	gender and birth_date 948 (Corporate and Fleet); monthly_income 1,470; closed_date 31; discontinued_date 56; engine_cc 12 (electric); delivery_date 969; financing_partner, down_payment_pct, tenor_months 3,868 (Cash).	Left as NULL, except financing_partner = "Not Used".
NULL, missing	email 423 (Retail); salesperson_name 164; lead_source 331.	email kept NULL; salesperson_name and lead_source = "Unknown".
Coded flags	marketing_opt_in, is_active (dealers, vehicles), is_trade_in, insurance_included are stored as 1 / 0.	Labels: Yes / No, Active / Inactive (dealers), Active / Inactive (vehicles).
transactions / csat_score	NULL in 5,890 rows: 969 not applicable (Cancelled, Pending) and 4,921 delivered deals with no response.	Keep NULL; add csat_category.
transactions / total_amount_idr	1,363 upper outliers (97.5% of Fleet transactions); formula check passes for every row.	Valid values; kept.
transactions / unit_price, discount	Unit price is above list price in 85.8% of rows (annual price increase); 4 transactions have a net price below vehicle cost.	Not errors; kept and flagged for analysis.

Example Approach:

1. Finding Duplicate Records

```
WITH standardized AS (
  SELECT customer_id, LOWER(TRIM(full_name)) AS name_key, registration_date,
    CASE WHEN CAST(phone AS STRING) LIKE '8%' THEN CONCAT('62', CAST(phone AS STRING))
        ELSE CAST(phone AS STRING) END AS phone_std
  FROM `project-nusantara-car.dataset_extraction.customers`
  WHERE customer_segment = 'Retail')
SELECT name_key, phone_std, COUNT(*) AS n_records,
      ARRAY_AGG(customer_id ORDER BY registration_date, customer_id) AS customer_ids
FROM standardized WHERE phone_std IS NOT NULL
GROUP BY name_key, phone_std HAVING COUNT(*) > 1;
```

Explanation: Retail customers with the same cleaned name and the same standardized phone number are grouped. The phone is standardized first because 24 pairs only match after the formats are aligned (raw numbers give 60 groups, standardized numbers give 84). Records without a phone are excluded, because GROUP BY would otherwise treat all NULLs as one group.

Result: 84 groups of two records; all 84 pairs also share the same birth_date and 82 the same city.

2. Finding NULL Values

```
SELECT COUNTIF(customer_id IS NULL) AS customer_id, COUNTIF(full_name IS NULL) AS full_name,
      COUNTIF(gender IS NULL) AS gender, COUNTIF(birth_date IS NULL) AS birth_date,
      COUNTIF(email IS NULL) AS email, COUNTIF(phone IS NULL) AS phone,
      -- ... one COUNTIF per column (13 columns in total)
FROM `project-nusantara-car.dataset_extraction.customers`;
```

Explanation: Each field is checked and its NULL values are counted; the same pattern was applied to all four tables. Only the columns listed in the summary above contain NULL values; every other column is complete.

3. Finding Non-Standardized Values

```
SELECT city, TRIM(INITCAP(city)) AS try_clean_city
FROM `project-nusantara-car.dataset_extraction.customers`
WHERE city <> TRIM(INITCAP(city)) LIMIT 1000;
```

Explanation: Each raw value is compared with its normalized version; if they differ, the data requires cleaning. The query returns 330 rows, and 149 spellings collapse into 36 cities.

4. Checking for Outliers and Anomalies in Numeric Values

```
WITH boundary AS (
  SELECT APPROX_QUANTILES(total_amount_idr, 100)[OFFSET(25)] AS q1,
       APPROX_QUANTILES(total_amount_idr, 100)[OFFSET(75)] AS q3
  FROM `project-nusantara-car.dataset_extraction.transactions`)
SELECT COUNT(t.total_amount_idr) AS total_count,
  COUNTIF(t.total_amount_idr < b.q1 - 1.5 * (b.q3 - b.q1)) AS lower_outliers,
  COUNTIF(t.total_amount_idr > b.q3 + 1.5 * (b.q3 - b.q1)) AS upper_outliers,
  MAX(b.q3 + 1.5 * (b.q3 - b.q1)) AS upper_bound
FROM `project-nusantara-car.dataset_extraction.transactions` t CROSS JOIN boundary b;
```

Explanation: Extreme values in total_amount_idr are found with the Interquartile Range (IQR) method (APPROX_QUANTILES is approximate, so counts may differ by a few rows).

Result with exact quartiles (Q1 255.6 million, Q3 440.8 million, upper bound 718.6 million IDR): 0 lower and 1,363 upper outliers (11.4%): 715 Fleet (97.5% of Fleet transactions), 426 Corporate, 222 Retail. They are legitimate bulk purchases (up to 15 units) and high-end vehicles, so they are kept. String anomalies are covered in step 3.

5. Data Validation

5.1 Datetime validation. The delivery date must not come before the transaction date.

```
SELECT * FROM `project-nusantara-car.dataset_extraction.transactions` WHERE delivery_date <
transaction_date;
```

Result: 0 rows (deliveries arrive 3 to 60 days after the transaction, median 31). Also 0 rows for: transaction before vehicle launch_date or after discontinued_date, before dealer opened_date or after closed_date, and before customer registration_date. Seven Returned transactions from September 2026 have delivery dates after the inspection date (2026-10-21 to 2026-11-18); they are immaterial and kept.

5.2 Calculation mismatch. total_amount_idr = quantity * (unit_price_idr - discount_idr); use <> to catch both directions.

```
SELECT transaction_id, total_amount_idr, (quantity * (unit_price_idr -
discount_idr)) AS comparison_total
FROM `project-nusantara-car.dataset_extraction.transactions`
WHERE total_amount_idr <> (quantity * (unit_price_idr - discount_idr));
```

Result: 0 rows. The formula holds for all 11,939 transactions, so the discount is a per-unit amount.

5.3 Business logic (1). unit_price_idr in transactions versus list_price_idr in vehicles.

```
SELECT t.transaction_id, v.list_price_idr, t.unit_price_idr, ROUND(t.unit_price_idr
/ v.list_price_idr, 4) AS price_ratio
FROM `project-nusantara-car.dataset_extraction.transactions` t JOIN `project-
nusantara-car.dataset_extraction.vehicles` v ON t.vehicle_id = v.vehicle_id
WHERE t.unit_price_idr <> v.list_price_idr;
```

Result: 10,239 rows (85.8%). The unit price is never below list and at most 1.076 times higher. The average ratio rises every year (1.000 in 2021 to 1.065 in 2026) while the vehicle table keeps one fixed list price. This is expected, not an error.

5.4 Business logic (2). Net selling price below vehicle cost (negative gross profit).

```
SELECT t.transaction_id, t.status, v.cost_idr, (t.unit_price_idr - t.discount_idr)
AS net_unit_price, (t.unit_price_idr - t.discount_idr - v.cost_idr) AS
gross_profit_per_uni
FROM `project-nusantara-car.dataset_extraction.transactions` t JOIN `project-
nusantara-car.dataset_extraction.vehicles` v ON t.vehicle_id = v.vehicle_id
WHERE v.cost_idr > (t.unit_price_idr - t.discount_idr);
```

Result: 4 transactions (T000568 Returned; T001748, T002363, T006817 Completed) with a loss of 0.3 to 2.0 million IDR per unit. Kept as examples of over-discounting for the analysis stage.

List of Data Fields:

1. Master Customers Table (10,690 rows, 13 fields)

1.A Field customer_id

Definition: The key identifier for each customer record (C00001 to C10690).

Issue: Unique (no NULL), but it identifies a registration, not necessarily a person: 84 people appear under two IDs (see 1.B). 881 customers (8.2%) have no transaction yet.

Handling: Never changed, because transactions reference it. A master_customer_id column is added instead.

1.B Field full_name

Definition: The customer's name: a person's name for Retail, a company name for Corporate and Fleet.

Issue: The same name is not the same person: 2,385 Retail names are shared by more than one record. Example: "Rangga Gunawan" has three records. C03079 and C05054 share email, phone, birth date and city (one person registered twice, the second time in capital letters); C07498 differs in all four (a different person).

> Real duplicates = same cleaned name and standardized phone (Retail, phone present): 84 pairs. All share the same birth_date and 82 the same city. Email is identical in 13 pairs, missing in one record in 44, different in 26 and missing in both in 1, so it is not part of the key. The strict test in the image (name, email and phone identical) finds only 14 of the 84.

43 Retail names are in capital letters (e.g. “DIAN UTOMO”); no name has stray spaces.

Row	customer_name	customer_email	customer_phone	jumlah_customers
1	rangga gunawan	rangga.gunawan19@yahoo.co.id	85376943032	2
2	indah wahyudi	indah.wahyudi96@icloud.com	89611132446	2
3	arif santoso	arif.santoso34@gmail.com	85240976602	2
4	indah wahyudi basri	indah.wahyudi54@gmail.com	81149071567	2
5	intan wijaya	intan.wijaya83@icloud.com	89688983011	2
6	teguh kurniawan permana	teguh.kurniawan9@gmail.com	89525266107	2
7	bagus halim	bagus.halim30@outlook.com	81258016668	2
8	jessica simanjuntak	jessica.simanjuntak1@gmail.com	81258991305	2
9	intan anggraini hutapea	intan.anggraini41@gmail.com	81345817176	2
10	teguh gunawan	*null*	89647481125	2
11	zainal hidayat hutapea	zainal.hidayat62@gmail.com	87759367351	2
12	rizky prakoso	rizky.prakoso39@yahoo.co.id	89921650830	2

Records with identical name, email and phone (strictest test).

Handling: No row is deleted and customer_id is not overwritten. master_customer_id is added: records with the same cleaned name and standardized phone share one master (the record with an email, then the earliest registration_date, then the lowest ID). Result: 10,606 unique customers; 56 transactions reference a non-master ID, so customer-level analysis must group by master_customer_id.

- Corporate and Fleet are never merged: 184 company names repeat, but 176 of those groups have different phones and registrations are 797 days apart on average.
- Known limitation: 6 further pairs (e.g. Putri Lubis, Mega Handayani, Budi Firmansyah) share name, birth date and city but have no phone to compare. One, Ahmad Permana (C09035, C10379), even shares the email. They stay separate under the name and phone rule.
- Names: INITCAP(TRIM()) for Retail; TRIM() only for companies, so PT and CV are not changed.

1.C Field gender

Definition: The customer's gender (M = Male, F = Female).

Issue: 948 NULL: exactly the 724 Corporate and 224 Fleet customers, for which gender does not apply. Retail has none (5,996 M, 3,746 F).

Row	gender	count_distinct
1	*null*	948
2	M	5996
3	F	3746

Handling: After coordination with the team, gender for companies is not replaced or edited.

1.D Field birth_date. The customer's date of birth. **Issue:** 948 NULL, the same Corporate and Fleet records (not applicable). Retail birth dates run from 1953 to 2005

Handling: None; NULL is left.

1.E Field email

Definition: The customer's email address.

Issue: 423 NULL, all Retail (4.3%). Existing emails are lower case with a valid format. 14 emails are shared by two records: 13 are the duplicate pairs in 1.B and the 14th is the Ahmad Permana pair.

Handling: Nothing is deleted. LOWER(TRIM()) as a safeguard; NULL stays NULL; email is only a tie-breaker when choosing the master record.

1.F Field phone

Definition: The customer's phone number.

Issue: Formats are inconsistent, and the half-and-half description is not accurate. In the CSV, 10,388 numbers start with "0" (e.g. 0812…), 24 start with "62" and 278 are empty. BigQuery stored the column as INT64 and dropped the leading zero, so they appear as 8… (10,388) or 62… (24, only 0.2%).

- All 24 numbers in the 62 format are the second registration of a duplicate pair, so standardization must happen before duplicate detection.

Handling: Cast to text; if the number starts with "8", add the prefix "62" (89775965062 becomes 6289775965062). Numbers already starting with "62" and NULL are unchanged. Every non-NULL result matches 628 followed by 9 to 10 digits.

1.G Field city

Definition: The city where the customer resides.

Issue: 149 spellings represent 36 cities; 330 rows (3.1%) are affected: 173 with leading or trailing spaces (e.g. " Bandung") and 157 with wrong capitalization (e.g. "BANDUNG", "surabaya"). No multiple spaces between words were found.

Row	city	count_distinct
1	Bandung	7
2	Bekasi	5
3	Jakarta Barat	2
4	Surabaya	4
5	Tangerang Selatan	2
6	Yogyakarta	3
7	BANDUNG	2
8	Balikpapan	161

Handling: Trim and convert to title case with INITCAP.

1.H Field province. The province where the customer resides. **Issue:** None. 22 provinces, no spacing or capitalization problems, and every city belongs to one province.

Handling: None.

1.I Field occupation

Definition: The customer's occupation. Eleven values: Karyawan Swasta, Wiraswasta, Profesional, PNS, Pengusaha, Freelancer, Guru/Dosen, TNI/Polri, Ibu Rumah Tangga, Pensiunan and Corporate.

Issue: "Pengusaha" (entrepreneur) and "Wiraswasta" (self-employed) are close in meaning, but their incomes differ clearly (median 46.0 vs 17.7 million IDR). "Guru/Dosen" and "Profesional" are not synonyms (median 7.2 vs 30.6 million IDR).

- All 948 Corporate and Fleet customers have occupation "Corporate", a placeholder for companies, so occupation analysis should use Retail only.
- Some combinations are atypical (about 80% of "Pensiunan" are under 50; 63.5% of "Ibu Rumah Tangga" are male). No correction is applied; occupation is a descriptive label.

Row	occupation	count_distinct
1	Corporate	948
2	Freelancer	585
3	Guru/Dosen	503
4	Ibu Rumah Tangga	496
5	Karyawan Swasta	2837
6	PNS	962
7	Pengusaha	909
8	Pensiunan	430
9	Profesional	990
10	TNI/Polri	382
11	Wiraswasta	1648

Handling: The values correspond to different monthly_income ranges, so they are left unchanged.

1.J Field monthly_income_idr. The customer's recorded monthly income. It is optional. **Issue:** 1,470 blank (13.8%): 948 Corporate and Fleet plus 522 Retail (5.4%). Retail incomes run from 1.9 to 190.1 million IDR (median 11.7 million).

Handling: After coordination with the team, blanks are at the customer's request or because company information is confidential. Not imputed.

1.K Field customer_segment. The customer classification entered by the salesperson at each dealership: Retail (individual, 9,742, always 1 unit), Corporate (company, 724, 1 to 5 units) and Fleet (bulk buyer, 224, 3 to 15 units). **Issue:** None. The segments differ strongly in value: Fleet is 6.1% of transactions but 33.0% of Completed revenue.

Handling: Kept as three segments.

1.L Field registration_date. The date when the customer registered. **Issue:** None. 2021-01-01 to 2026-09-27, no NULL, and no customer transacts before registering. A repeat registration comes 13 to 1,341 days after the first.

Handling: None.

1.M Field marketing_opt_in. Whether the customer allows the dealer to contact them for follow-ups or promotions. **Issue:** No NULL, but stored as a code: 1 (6,999) and 0 (3,691). **Handling:** 1 = "Yes", 0 = "No", other = "Unknown".

2. Master Dealers Table (32 rows, 14 fields)

Field	Definition	Issue and handling
2.A dealer_id	Key identifier of each dealer (D001 to D032).	**Issue:** None. Unique; no orphans in transactions.
2.B dealer_name	The dealer's name, "Nusantara Motor <city> - <area>".	**Issue:** None. 32 unique names.
2.C dealer_type	Business type: Main Dealer (10), Authorized Dealer (17), Satellite Showroom (5).	**Issue:** None.
2.D city	City where the dealer is located.	**Issue:** None.
2.E province	Province where the dealer is located.	**Issue:** None.
2.F region	Dealer region: Jabodetabek (11), Jawa (8), Sumatera (7), Kalimantan (3), Sulawesi (2), Bali & Nusa Tenggara (1).	**Issue:** No formatting problem. Region is not a grouping of provinces: Jabodetabek includes Jawa Barat dealers (Bogor, Depok, Bekasi) and Jawa also has Jawa Barat dealers (Bandung). **Handling:** None.
2.G address	Address of the dealer location.	**Issue:** None.
2.H phone	The dealer's phone number.	**Issue:** Same loading problem as customers: the CSV has 0 plus area code (0657194391), BigQuery INT64 dropped the 0 (9 digits). **Handling:** Add prefix "62" (62657194391). No number starts with 62.
2.I manager_name	Manager responsible for the dealer.	**Issue:** None. Every dealer has a different manager.
2.J opened_date	Date the dealer opened (2008-10-30 to 2026-02-01).	**Issue:** None. No transaction before the opening date.
2.K closed_date	Date the dealer closed.	**Issue:** 31 NULL (dealers still open). Only D024 (Pekanbaru) is closed (2024-11-30); no transaction after it. **Handling:** None; the NULL is meaningful.
2.L is_active	Whether the dealer still operates; inactive when closed_date is populated.	**Issue:** Stored as a code (1 = 31, 0 = 1) and consistent with closed_date. **Handling:** 1 = "Active", 0 = "Non Active".
2.M monthly_sales_target_units	Monthly unit sales target (5 to 27).	**Issue:** None.
2.N showroom_area_sqm	Showroom floor area in square meters (301 to 983).	**Issue:** None.

3. Master Vehicles Table (62 rows, 16 fields)

Field	Definition	Issue and handling
3.A vehicle_id	Key identifier of each vehicle (V001 to V062).	**Issue:** None. Unique; no orphans in transactions.
3.B brand	Brand (13 brands).	**Issue:** None.
3.C model	Model (56 models).	**Issue:** None.
3.D variant	Model variant.	**Issue:** None. Every row is a unique variant.
3.E body_type	SUV (25), MPV (16), Hatchback (12), Pickup (4), Sedan (4), Van (1).	**Issue:** None.
3.F fuel_type	Petrol (38), Electric (12), Hybrid (7), Diesel (5).	**Issue:** None.
3.G transmission	CVT (29), Automatic (23), Manual (9), AMT (1).	**Issue:** None.
3.H engine_cc	Engine displacement (998 to 2,494 cc).	**Issue:** 12 NULL: all electric vehicles (no combustion engine). **Handling:** None; not applicable.
3.I seating_capacity	Number of seats (2, 4, 5 or 7).	**Issue:** None.
3.J assembly_origin	Where the vehicle is assembled: Local Assembly (46, in Indonesia) or Imported (16, produced abroad).	**Issue:** None.
3.K model_year	Model year (2022 to 2027).	**Issue:** V061 (Honda Brio e:N1) and V062 (BYD Sealion 7) have model_year 2027 and launch dates in November 2026: pre-launch models with no transactions. model_year can differ from the launch year (V001: launched 2021, model_year 2023). **Handling:** None; kept.
3.L list_price_idr	Official list price (150 million to 1.6 billion IDR).	**Issue:** One fixed price per vehicle, while transaction unit prices rise over time (see 4.H). **Handling:** None; reference only.
3.M cost_idr	Cost of supplying the vehicle to the dealer.	**Issue:** One fixed value per vehicle (no cost history), 7.3% to 13.9% below list price. 4 transactions are sold below it (see 4.I). **Handling:** None; joined when margin is analyzed.
3.N launch_date	Date the vehicle went on sale (one date per vehicle, not per dealer; 2021-01-01 to 2026-11-15).	**Issue:** None. No transaction before the launch date.
3.O discontinued_date	Date the vehicle stopped being sold.	**Issue:** 56 NULL (still on sale). Six vehicles are discontinued and none sells after that date; two transactions (T006344, T007658) fall on the discontinued date itself, the last day of sale. **Handling:** None; the NULL is meaningful.
3.P is_active	Whether the vehicle is still on sale; inactive when discontinued_date is populated.	**Issue:** Stored as a code (1 = 56, 0 = 6) and consistent with discontinued_date. **Handling:** 1 = "Active", 0 = "Inactive".

4. Transactions Table (11,939 rows, 20 fields)

Field	Definition	Issue and handling
4.A transaction_id	Key identifier of each sales transaction.	**Issue:** None. 11,939 unique values.
4.B transaction_date	Date of the transaction (2021-01-01 to 2026-09-27).	**Issue:** None. No NULL.
4.C customer_id	Reference to customers.	**Issue:** No orphans (9,809 distinct customers). 56 transactions reference the second registration of a duplicate customer. **Handling:** Not changed; group by master_customer_id for customer-level analysis.
4.D vehicle_id	Reference to vehicles.	**Issue:** None. No orphans; every sale is within the vehicle's launch and discontinued dates.
4.E dealer_id	Reference to dealers.	**Issue:** None. No orphans; every sale is within the dealer's opened and closed dates.
4.F salesperson_name	Salesperson responsible for the transaction.	**Issue:** 164 NULL (1.4%), spread over 29 of the 32 dealers and all statuses; no blank strings. **Handling:** Replace NULL with "Unknown".
4.G quantity	Number of vehicles in the transaction (1 to 15).	**Issue:** None. 87.9% are single units. Retail buys 1, Corporate 1 to 5, Fleet 3 to 15.
4.H unit_price_idr	Agreed price of one unit, before discount.	**Issue:** Differs from vehicles.list_price_idr in 85.8% of rows: never lower, at most 1.076 times higher, rising from 1.000 (2021) to 1.065 (2026) on average because of annual price increases. **Handling:** None; not an error.
4.I discount_idr	Discount per vehicle unit (not per transaction).	**Issue:** 3,479 transactions (29.1%) have no discount; the average is 3.1% of the unit price when given. In 4 transactions the net price is below vehicle cost (negative gross profit). **Handling:** None; flagged for analysis.
4.J total_amount_idr	quantity * (unit_price_idr - discount_idr); present for every status, so it is revenue only when status is Completed.	**Issue:** The formula holds in all rows. 1,363 upper outliers (see step 4), which are valid. **Handling:** None. Revenue = Completed only (5.61 of 6.30 trillion IDR).
4.K payment_method	Leasing (4,951), Cash (3,868), Bank Credit (3,120).	**Issue:** None.
4.L financing_partner	Partner for Leasing (BCA Finance, WOM Finance, TAF, Adira Finance, ACC, Mandiri Tunas Finance) and Bank Credit (CIMB Niaga, BNI, BCA, BRI, Mandiri).	**Issue:** 3,868 NULL: exactly the Cash transactions. Every financed transaction has a partner. **Handling:** Replace NULL with "Not Used".
4.M down_payment_pct	Down payment percentage (15% to 40%).	**Issue:** 3,868 NULL (Cash). **Handling:** None; not replaced with 0, which would distort averages.
4.N tenor_months	Payment months (12, 24, 36, 48 or 60).	**Issue:** 3,868 NULL (Cash). **Handling:** None; not replaced with 0.
4.O lead_source	Marketing channel: Walk-in, Website, Exhibition, Referral, Instagram, Telemarketing, Marketplace.	**Issue:** 331 NULL (2.8%); no blank strings. **Handling:** Replace NULL with "Unknown".

Field	Definition	Issue and handling
4.P is_trade_in	Whether a vehicle was traded in.	**Issue:** No NULL, but a code: 1 (1,434; 12.0%) and 0. **Handling:** 1 = "Yes", 0 = "No".
4.Q insurance_included	Whether insurance is included.	**Issue:** No NULL, but a code: 1 (7,204; 60.3%) and 0. **Handling:** 1 = "Yes", 0 = "No".
4.R status	Completed (10,655), Cancelled (760), Returned (315), Pending (209).	**Issue:** No formatting problem. For analysis, Pending is a booked sale that is not revenue yet; Cancelled and Returned are excluded. Pending occurs for all payment methods (Leasing 91, Cash 63, Bank Credit 55). **Handling:** None; all statuses are kept and filtered in analysis (revenue = Completed).
4.S delivery_date	Date the vehicle was delivered.	**Issue:** 969 NULL: exactly the Cancelled (760) and Pending (209) transactions. Delivery is 3 to 60 days after the transaction and never before it. **Handling:** None; the NULL is meaningful.
4.T csat_score	Customer satisfaction score, 1 to 5.	**Issue:** 5,890 NULL (49.3%): 969 not applicable (Cancelled, Pending) and 4,921 delivered deals with no response. The average among respondents is 3.99. **Handling:** Keep NULL (not 0). Add csat_category: Satisfied (4 to 5), Netral (3), Dissatisfied (1 to 2), N/A, No Response.